

3D Modeling Tools

There are several software available that help you create your own 3D models. This month's EFY-CD packs some Open Source software for 3D modeling which can be installed and used free of cost for as long as you need them

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Three-dimensional (3D) modeling is the procedure of developing 3D models using specialised software. We can say that it is the process of developing a mathematical representation of any 3D surface of the object by using computer application.

Nowadays 3D modeling is being used for many different types of activities. Be it video game or medical science, computer-aided design (CAD) programs to create 3D models of new devices, vehicles and structures too.

There are several software available that help you create your own 3D models. Below are few of the Open Source software for 3D modeling which can be installed and used free of cost for as long as you need them.

BRL-CAD

BRL-CAD is a powerful cross-platform Open Source solid modeling system.

What's Special?

It has tools for interactive geometry editing, high-performance ray-tracing for rendering and geometric analysis, and image and signal processing. It also includes a system performance analysis benchmark suite tool and libraries for robust geometric representation. The strength of the package lies in its ability to build and analyse realistic models of complex objects using a relatively small set of primitive shapes.

Distributor: Being free and Open Source, this software can be downloaded freely from the website of the software. Home

page: <http://brlcad.org/>

Download URL: <http://brlcad.org/d/download>

Platforms Supported: Microsoft Windows, GNU/Linux, MacOS X, Solaris, IRIX and BSD

FreeCAD

FreeCAD is an Open Source 3D CAD/CAE program, based on OpenCascade, QT and Python.

What's Special?

It is a basic 3D CAD with advanced motion-simulation capabilities. It features key concepts like macro recording, workbenches, ability to run as server and dynamically loadable application extensions. It is suitable for anyone interested in learning 3D CAD and motion simulation for free before using more sophisticated packages. Users can create and manipulate assemblies of parts. The parts are simple 3D solids, which can be connected by joints, constraints, contacts, motors, actuators, springs, dampers, forces, torques or gravity.

Distributor: Being free and Open Source, this software can be downloaded freely from the website of the software.

Home page: <http://free-cad.sourceforge.net>

Download URL: <http://sourceforge.net/projects/free-cad/files/>

Platforms Supported: Microsoft Windows, GNU/Linux and MacOS X

AutoQ3D CAD

AutoQ3D CAD helps you create 2D and 3D technical drawings and sketching designs as well.

What's Special?

AutoQ3D CAD is easy to use and fast, which makes it perfect for any beginner or novice to get into technical drawing without the hassle and complexity of other applications.

Distributor: It can be freely downloaded from website.

Home page: www.autoq3d.com

Download URL: www.autoq3d.com/products/open-source-3d-cad.html

Platform Supported: Microsoft Windows

K-3D

K-3D is a free 3D modeling and animation software.

What's Special?

It is a versatile and powerful tool for artists that allows flexible plugins with a visualisation pipeline architecture. Its interface uses your platform's look-and-feel, and is consistent with the applications you already know. Its advanced undo/redraw system ensures you can always go back, and the undo tree means you can always go forward, even where other programs lose the ability to redo. Artists will find K-3D easy to understand, and professionals feel right at home. The application is written in C++ with platform independence in mind.

Distributor: Being free and Open Source, this software can be downloaded freely from the website of the software.

Home page: www.k-3d.org

Download URL: www.k-3d.org/downloads
Platforms Supported: Microsoft Windows, MacOS X, GNU/Linux and FreeBSD ●

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